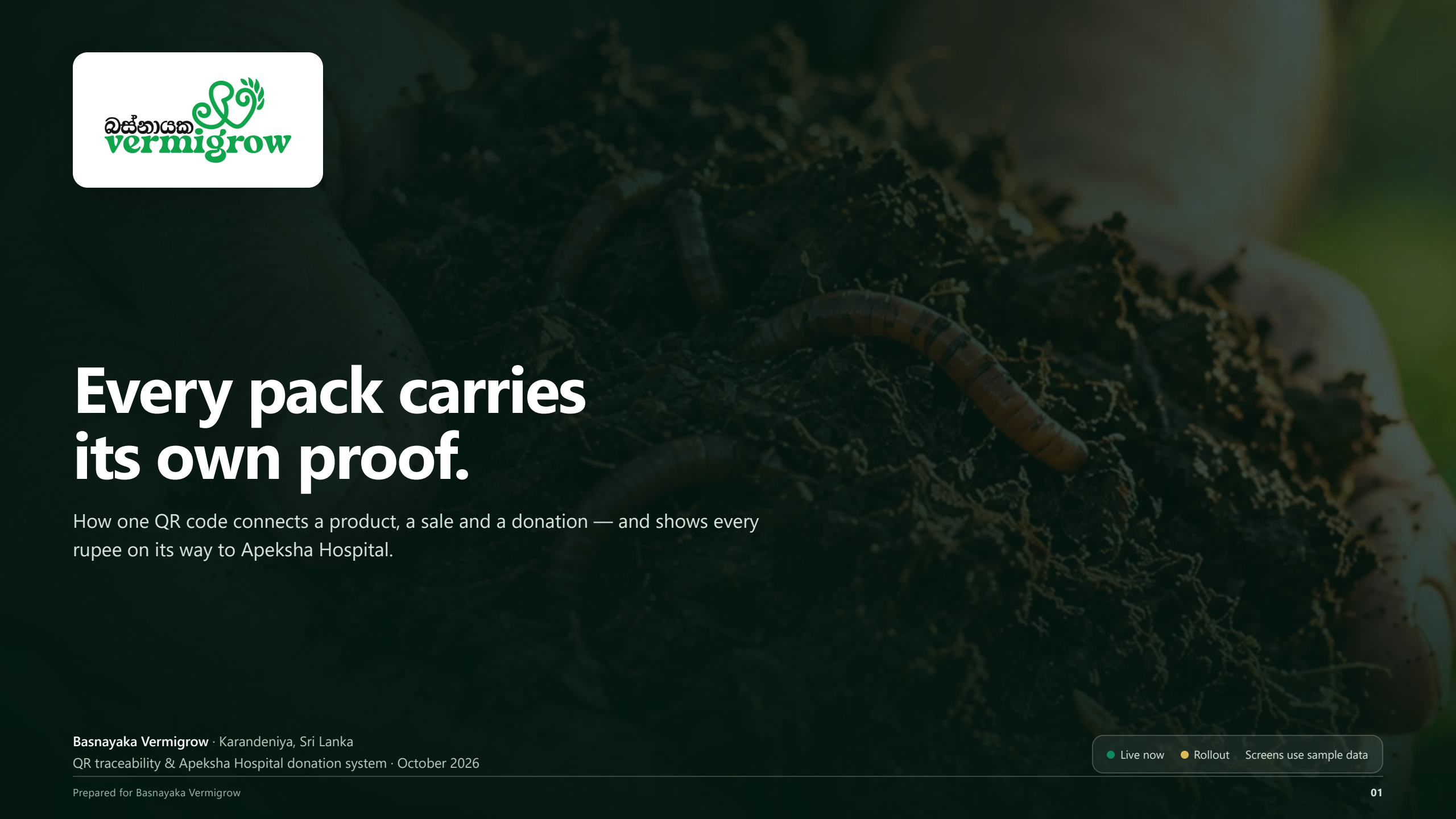


Every pack carries its own proof.

How one QR code connects a product, a sale and a donation — and shows every rupee on its way to Apeksha Hospital.

Basnayaka Vermigrow · Karandeniya, Sri Lanka
QR traceability & Apeksha Hospital donation system · October 2026

Live now · Rollout · Screens use sample data

THE WHOLE PROCESS

LIVE NOW

From a catalogue item to a donor’s name on the wall

One continuous loop. The QR code is only the hinge — the system carries the unit’s identity, its sale, its donation and the hospital record through every step.

01

Add the item

A product or a service, with its price and donation per unit.

OUTPUT
A sellable item and its donation rate

→

02

Generate codes

One unique QR code for every pack or workshop seat.

OUTPUT
A batch of unique pack codes

→

03

Print & label

A sticker sheet PDF that fits any paper size.

OUTPUT
A printed, page-ready sticker sheet

→

04

Dispatch

Record which retailer holds which codes.

OUTPUT
A dispatch record per retailer

05

Sell

The buyer scans, or staff record the sale for them.

OUTPUT
A sale — scanned or recorded

→

06

Detect the sale

First scan marks the unit sold; vouchers redeem.

OUTPUT
A sold pack; a redeemed seat

→

07

Count it once

The donation enters the ledger exactly one time.

OUTPUT
One ledger entry, one time

→

08

Show & send

The wall updates; the money is reconciled to Apeksha Hospital.

OUTPUT
A public total and a donor receipt

STEP 01 · CATALOGUE

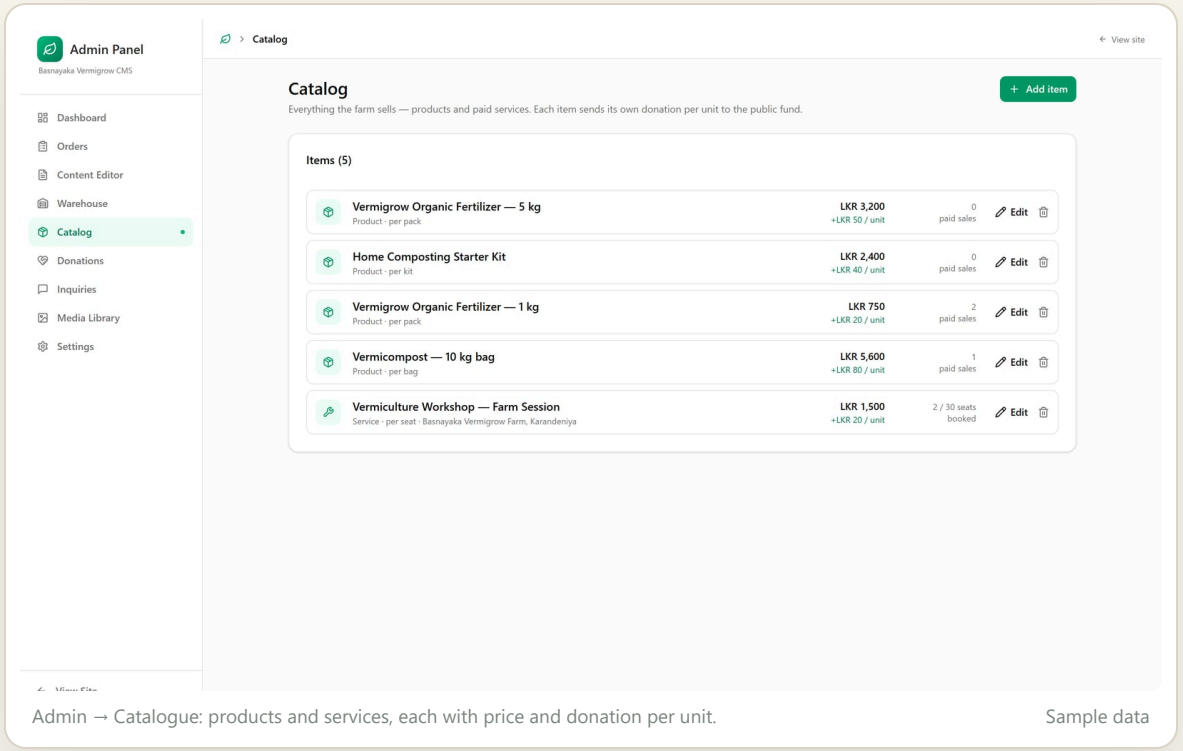
LIVE NOW

Add or customise any product — or any service

Everything sellable is created in one place. A product is a physical pack; a service is a paid seat at a workshop. Both carry their own donation rate.

- **One catalogue, two kinds.** Products and services share the same screen, but behave differently: a service can carry a date, a venue and a seat cap.
- **Price and donation per unit** are set per item. The donation is copied onto each sale at entry, so changing the rate tomorrow never rewrites yesterday's history.
- **Photography, description and availability** are managed here too — the public order page reads this catalogue directly.
- **Seat capacity is enforced** for workshops: bookings cannot exceed the cap, and remaining seats are shown to the customer.

> **What the client controls:** the product line, the donation rate and what appears online — without calling a developer.



Admin → Catalogue: products and services, each with price and donation per unit. Sample data

STEP 02 · UNIQUE CODES

LIVE NOW

Every pack gets an identity, not just a sticker

A batch of codes is generated for a chosen product. Each code belongs to exactly one physical pack — or one workshop seat — for its whole life.

VG-7F3K9Q

fixed prefix | 6 characters | 32-letter alphabet

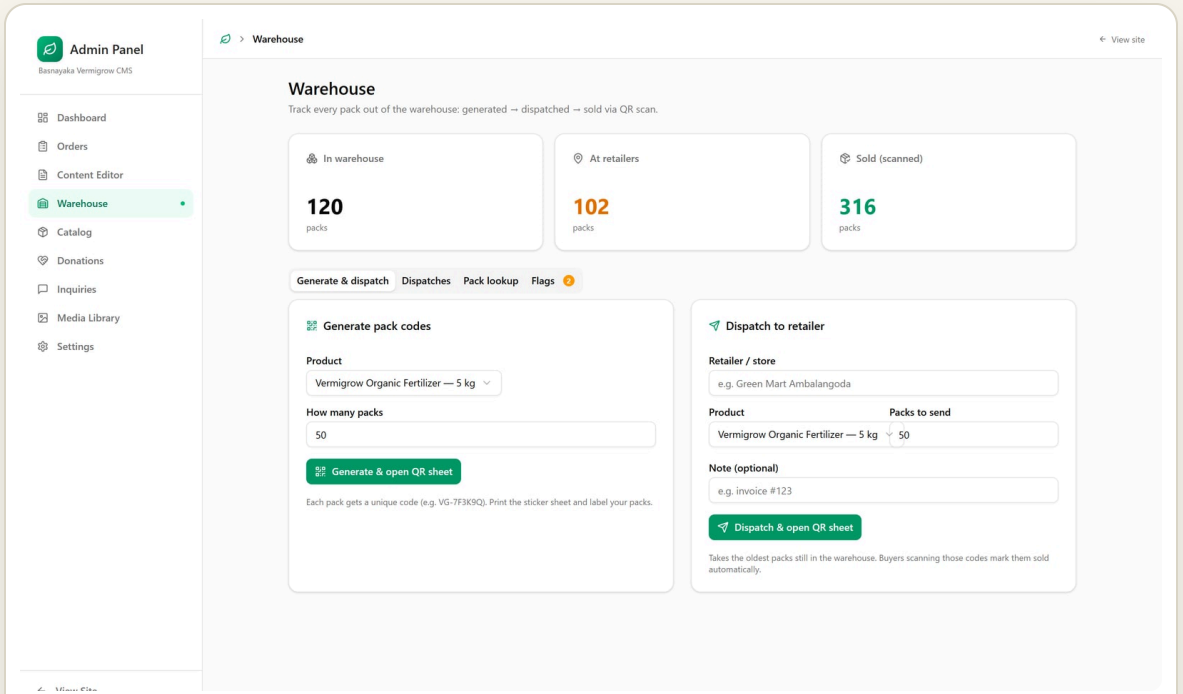
32^6

≈ 1.07 billion combinations — collisions are negligible and still blocked by a unique database constraint.

5,000

codes per batch, enough for a full production run.

- **No 0, O, 1 or I** in a code — it stays readable when handwritten, dictated over the phone, or read from a damaged label.
- **The code is the record key** for the pack: its status, scan history, dispatch and donation all hang off it.
- **Not guessable by design.** A code is short because it is printed in public; the private receipt uses a separate, unguessable token.



Admin → Warehouse: generate a batch, then print the sheet or dispatch the codes to a retailer. Sample data

STEP 03 · PRINT & LABEL

LIVE NOW

A sticker sheet that fits the printer you already have

The same layout engine drives the on-screen preview, the browser print and the downloadable PDF. What you see is what prints.

- **Any common paper:** A4, A5, A6, Letter, Legal — or a custom width and height in millimetres.
- **Adjustable to the label stock:** orientation, page margin, QR size (42 mm default) and the gap between stickers.
- **Each sticker carries** the brand, the QR, the code and the product name — nothing a shopkeeper has to interpret.
- **Scale:** a server-side PDF handles up to 5,000 codes (500 pages). The browser preview is capped at 200 for speed.
- **Dispatch is recorded:** the system knows which codes went to which retailer, and removing a dispatch returns those packs to stock.

> **Label stock, not a special printer.** A normal office printer and ordinary sticker sheets are enough.



Sticker anatomy — brand · QR · code · product

Actual sheet: 12–60 per page

STEP 04 · THE SALE

LIVE NOW

Two ways to sell — because most buyers will not scan

The QR is the direct path, not the only path. A sale recorded by staff is still a tracked sale, with the same donation and a receipt sent to the buyer.

The buyer scans the pack

The direct, self-service path

1. The QR opens the donation page with the pack code already filled in.
2. The first scan marks that pack **sold** and writes the donation to the ledger.
3. The buyer can add their name to the public wall; the amount is taken from the pack, never typed by the buyer.

WHAT THE BUYER SEES

Pack verified — LKR 20 added to the Apeksha Hospital fund. Thank you!

Recorded as	**A scan event**
Donation taken from	**The pack code**
Name on the wall	**Optional, self-entered**

Best for: customers who want to see the record themselves, and for partner retailers where no staff member is involved.

Staff record the sale

The common shop-floor path

1. A member of staff enters the sale in the admin, with the buyer's phone number.
2. The system texts a private receipt link to the buyer — no app, no login.
3. The sale carries its donation into the ledger; the QR on the pack stays optional.

SMS TO THE BUYER

Basnayaka Vermigrow: your order added LKR 120 to Apeksha Hospital. Your receipt: .../r/••••••

Recorded as	**A sale, by staff**
Receipt delivered	**Private SMS link**
QR scan needed	**No**

Built around the real counter: a walk-in paying cash without a smartphone is still a tracked donation.

STEP 05 · THE SYSTEM DETECTS THE SALE

LIVE NOW

One scan changes the state — and only once

A pack and a workshop seat move through different lifecycles, but the same rule protects both: the first event is the one that counts.

A PHYSICAL PACK

In warehouse → Dispatched → Sold · first scan

A WORKSHOP SEAT (VOUCHER)

Issued when paid → Redeemed at the workshop

Every scan is logged
A ScanEvent is written even when it is a repeat.

Repeat scans are visible
They are flagged for review and never add money.

Vouchers carry LKR 0
The paid sale carries the donation; the voucher cannot count it twice.

Packing ≠ paying
Fulfilment stages are separate from the money state.

WHAT A SCAN RECORDS

Pack code VG-7F3K9Q	**Status change** dispatched → sold
Scan history time, device, repeat count	**Donation** entered in the ledger once
Anomaly signal same code scanned again	**No second payment** the total does not move

First event wins
Every later event is evidence, not income

Every scan leaves evidence. Only the first one leaves a donation.

Repeat scans are shown to staff for review — they never change the public total.

STEP 06 · THE MONEY PATH

LIVE GATEWAY ROLLOUT

Three ways money arrives. One ledger records the gift.

Cash and bank transfer are live today. The online gateway is designed and next; it is labelled as rollout, not claimed as live.

Online payment

ROLLOUT

Payments.lk hosted checkout

- Order total is built from our own database, never from the browser.
- The customer pays on the provider's page.
- Only the signed webhook can confirm payment.
- Confirmation settles the order and writes the ledger in one transaction.

Buyer sees	**Hosted checkout**
System stores	**Provider reference**
Confirms payment	**Signed webhook**
Currency	**LKR, locked server-side**

When live: the donation appears on the wall as soon as payment is confirmed — no staff step.

Bank transfer

LIVE

Recorded by the farm office

- Customer transfers and sends the slip or reference.
- Staff confirm receipt and record the bank reference.
- The order becomes paid; the donation is written once.
- The reference is the reconciliation key against the statement.

Buyer sees	**Bank slip**
System stores	**Bank reference**
Confirms payment	**Farm office**
Statement match	**By bank reference**

Today: a human confirms the transfer. The system keeps the reference, the receiver and the amount.

Cash in hand

LIVE

Counter and field sales

- Staff record the cash sale, or the seller records it later.
- Cash is deposited to the Apeksha Hospital donation account.
- The deposit is reconciled against the ledger.
- No buyer needs a phone or a bank account.

Buyer sees	**SMS receipt**
System stores	**Seller + deposit ref**
Confirms payment	**Deposit reconciliation**
Collector	**Named staff member**

Today: the system records the promise; the physical deposit remains a finance step, checked against the ledger.

One append-only donation ledger

Every channel feeds the same total — the number on the public wall.

The ledger records what was raised. Moving the money into the hospital account is a separate, reconcilable step — manual today, automatable later.

STEP 07 · INTEGRITY

LIVE NOW

Counted once, never twice — enforced by the database

The donation total is the one number the client, the buyer and the hospital all see. It is protected structurally, not by remembering to be careful.

Append-only ledger

One row per donation-bearing event. The public total is a single query over it, whatever the source.

Unique source key

Channel + record id is unique. A retried scan, a double-clicked “mark paid” or a replayed event is a no-op.

Snapshotted amounts

The donation is copied at sale time. Editing today’s rate cannot restate yesterday’s figure.

Cancelled, not deleted

A cancelled order reduces the total but the ledger row stays, so history remains auditable.

Money state ≠ fulfilment

Packing or delivering an order is a separate field; it cannot move money as a side effect.

Vouchers are excluded

A service voucher writes LKR 0; the paid sale carries the donation. Two independent guards.

HOW EVERY SOURCE REACHES THE WALL

Scanned pack channel: pack	→	**Ledger entry** unique by pack id
Recorded sale channel: sale	→	**Ledger entry** unique by sale id
Direct gift channel: direct	→	**Ledger entry** unique by gift id
Replayed webhook same event id	→	**Rejected** the unique key is already used

One public total

Only entries marked paid are summed

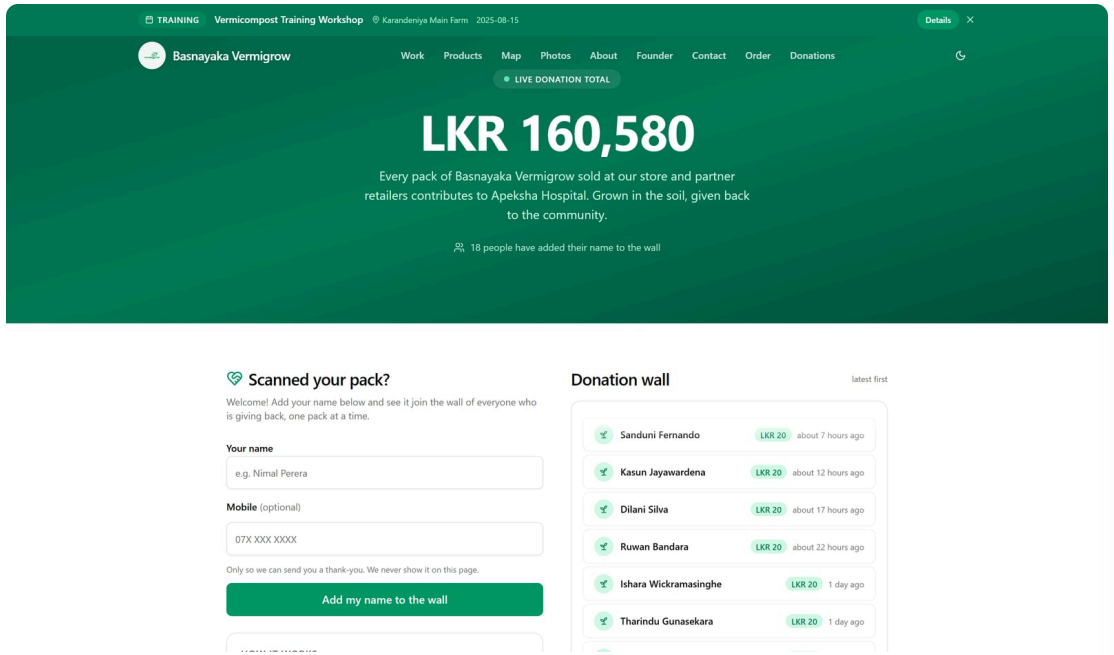
A duplicate is rejected by the database, not by a rule someone has to remember.

STEP 08 · WHAT THE DONOR SEES

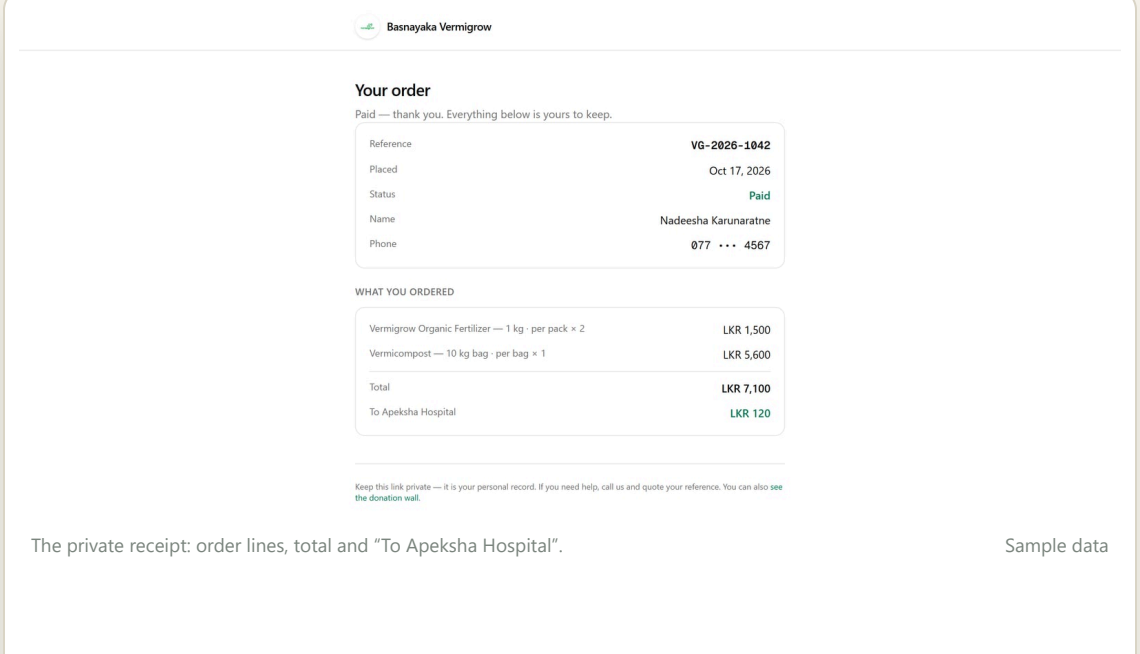
LIVE NOW PERSONAL TOTALS

The name, the amount, and a receipt they can keep

The public wall shows the running total and the latest names. A private receipt shows one buyer exactly what their purchase sent to Apeksha Hospital.



The live wall: total, story and latest donor names. Sample data



The private receipt: order lines, total and “To Apeksha Hospital”. Sample data

- **Refreshes every 10 seconds** — a sale at a retailer appears while the page is open.
- **Phone numbers are never shown** — not on the wall, not in the public feed.
- **Every purchase has a record** — order, pack, gift or recorded sale.

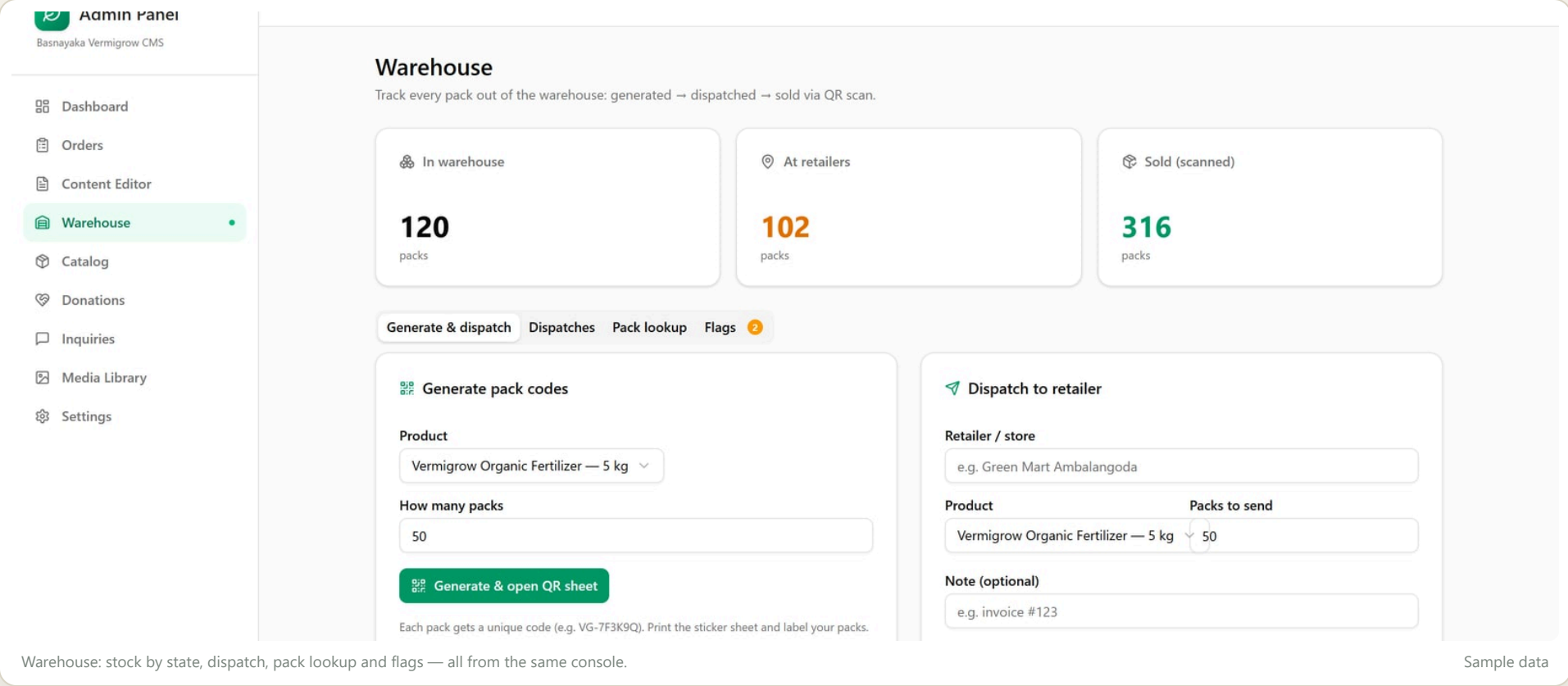
- **Amount is server-attributed** from the scanned pack, never typed by the visitor.
- **Receipt links are private**: unguessable, not indexed, expirable and revocable.
- **Lifetime donor totals** across purchases are designed but not live yet — marked rollout.

STEP 09 · OVERSIGHT

LIVE NOW

The back office sees every pack and every rupee

The same system that sells also answers the operational questions: what is in stock, what is sitting at a retailer, what was scanned twice, and what has actually been paid.



Warehouse: stock by state, dispatch, pack lookup and flags — all from the same console.

Sample data

- **Warehouse.** In stock, at retailers and sold — with a per-retailer list of packs still unsold after 30 days.
- **Orders.** Placed → confirmed → packed → dispatched → delivered, separate from payment state.
- **Anomalies.** Codes scanned more than once are surfaced for review; the money never changes.
- **Payments.** Method, bank reference, receiver, and one receipt that can settle a multi-line order.
- **Pack lookup.** Enter a code and see its product, retailer, status and full scan history.
- **Ledger & access.** Channel totals, the exact wall total — and every admin route authenticated.

STATUS · OCTOBER 2026

LIVE NOW ROLLOUT

What is live today — and what is deliberately next

No stage is blurred. Everything below is either running in the system or clearly marked for rollout.

Live now — Running end to end

- ✓ Product & service catalogue
- ✓ Sticker PDF & browser print
- ✓ Scan-to-sold detection
- ✓ SMS receipt links
- ✓ Private per-purchase receipts
- ✓ Cash & bank-transfer recording
- ✓ Unique pack / seat codes
- ✓ Dispatch to retailers
- ✓ Staff-recorded sales
- ✓ Donation ledger & wall
- ✓ Service vouchers
- ✓ Warehouse / order oversight

The full loop — item to sale to ledger to donor record — is working. Sample data is used only for the screens in this deck.

Rollout — Designed, not live

- ◷ Payments.lk hosted checkout
- ◷ Automatic online payment confirmation
- ◷ Personal lifetime donor totals
- ◷ Automated hospital deposit reconciliation

The gateway work is planned in detail and awaiting approval. Until it ships, no online card payment is taken.

EVIDENCE BEHIND "LIVE"

— A generated code is printed, dispatched and scanned — the same pack moves from stock to sold.

— A recorded sale produces an SMS receipt and exactly one ledger entry.

— The public total is computed from the ledger, not typed in by hand.

WHAT IS STILL MISSING FOR ROLLOUT

— A Payments.lk account and signed webhook for online confirmation.

— Donor identity linking for lifetime totals across purchases.

— A signed monthly deposit report from the bank to the hospital account.

THE TEST WE APPLIED **Live** means the complete path works end to end in the current build — data, money and donor record. **Rollout** means any step still needs building, approval or a provider account. No feature is shown as live on the strength of a mock-up.

PROPOSED ROLLOUT

FOR APPROVAL

A three-phase path, in the order that protects the money

Phase 1 makes the current system operational in the shop. Phase 2 adds the gateway. Phase 3 closes the loop between the ledger and the hospital account.

PHASE 1 · NOW

Operate what is live

Put the current build to work before adding new money paths.

- Deploy the current build and verify it in production.
- Confirm Apeksha Hospital wording on every public page.
- Print the first sticker batch; label and dispatch stock.
- Train counter staff on seller-assisted sales and SMS receipts.
- Agree the public donor-name policy.

DONE WHEN

The first labelled batch is on a shelf and its sales appear in the ledger.

Outcome: every pack on the shelf is traceable today.

PHASE 2 · GATEWAY

Add online payment

One hosted checkout, one signed webhook, one money path.

- Approve the Payments.lk decisions and security prerequisites.
- Open the provider account and sandbox keys.
- Build checkout + signed webhook; test success, decline and expiry.
- Settle the order and write the ledger in one transaction.
- Decide failed-payment and refund handling.

DONE WHEN

One real LKR 1 test payment confirms, settles and lands in the ledger.

Outcome: a card payment confirms the donation without staff.

PHASE 3 · DONOR & FINANCE

Close the loop

Make the donor record complete and the hospital deposit auditable.

- Personal donation history for returning buyers.
- Monthly hospital deposit report from the ledger.
- Reconcile bank transfer, cash and gateway totals.
- Agree who signs off each deposit.
- Publish the monthly hospital total.

DONE WHEN

The first monthly hospital deposit is signed off against the ledger.

Outcome: every rupee raised is traceable to a hospital deposit.

WHAT WE NEED • Hospital donation account details • Payments.lk account & keys • Who confirms bank transfers • Approval of the gateway scope • A launch date for Phase 1

THE SEQUENCING RULE No new money path goes live until the previous one reconciles cleanly — from the sale, to the ledger, to a signed hospital deposit.

සිංහල සාරාංශය

දැන් ක්‍රියාත්මකයි ඊළඟ අදියර

එක් QR කේතයකින් එක් රෝහල් පරිත්‍යාගයක්

Basnayaka Vermigrow නිෂ්පාදනයක් හෝ වැඩමුළු ආසනයක් විකිණීමේ සිට Apeksha රෝහලට ලැබෙන පරිත්‍යාගය දක්වා සම්පූර්ණ ක්‍රියාවලිය.

1. නිෂ්පාදනය සහ QR කේතය. සෑම විකුණුම් ඒකකයකටම — එක් පැකට්ටුවකට හෝ එක් වැඩමුළු ආසනයකට — අද්විතීය QR කේතයක් ලැබේ. කේතය පැකට්ටුවේ ඇති ස්ටිකරයේ මුද්‍රණය කර, අවශ්‍ය ප්‍රමාණයේ කඩදාසියකට PDF ලෙස බාගත හැක.

2. විකුණුම හඳුනා ගැනීම. පාරිභෝගිකයා QR කේතය ස්කෑන් කළ විට එම ඒකකය විකුණුණු බව පද්ධතිය ස්වයංක්‍රීයව හඳුනා ගනී. QR ස්කෑන් නොකරන පාරිභෝගිකයින් සඳහා කාර්ය මණ්ඩලය විකුණුම ලියාපදිංචි කර, SMS මගින් පෞද්ගලික රිසිට්පතක් යවයි. QR නොමැතිව විකුණුමක් සිදුවුවත් පරිත්‍යාගය නැති නොවේ.

3. පරිත්‍යාගය එක් වරක් පමණක්. සෑම විකුණුමකින්ම Apeksha රෝහලට ලැබෙන මුදල පරිත්‍යාග ලෙජරයේ එක් වරක් පමණක් ලියාපදිංචි වේ. එකම කේතය නැවත ස්කෑන් කළත් මුදල දෙවරක් එකතු නොවේ. අවලංගු කළ ඇණවුමක් මුළු එකතුවෙන් ඉවත් වන නමුත් එහි ඉතිහාසය ඉතිරි වේ.

4. පාරිභෝගිකයාට පෙනෙන දේ

රෝහල් අරමුදල් පුවරුව සජීවීව යාවත්කාලීන වේ — මුළු පරිත්‍යාග මුදල සහ පරිත්‍යාගිකයන්ගේ නම්. තම මිලදී ගැනීමට අදාළ පෞද්ගලික රිසිට්පතේ, Apeksha රෝහලට ගිය නිශ්චිත මුදල දැකිය හැක. දුරකථන අංක කිසිවිටෙකත් ප්‍රසිද්ධියේ නොපෙන්වයි.

5. මුදල් ගෙවීම් සහ රෝහල

දැනට මුදල් සහ බැංකු මාර්ගයෙන් ගෙවීම් ක්‍රියාත්මකයි; කාර්ය මණ්ඩලය ලැබීම තහවුරු කර ලෙජරයට එක් කරයි. අන්තර්ජාල ගෙවීම් ද්වාරය (Payments.lk) ඊළඟ අදියරේදී එකතු වේ. එවිට ගෙවීම තහවුරු වූ වහාම පරිත්‍යාගය ස්වයංක්‍රීයව ලියාපදිංචි වේ. සියලු පරිත්‍යාග Apeksha රෝහලේ පරිත්‍යාග ගිණුමට නැන්පත් කර මාසිකව ලෙජරය සමඟ සසඳා බැලේ.

එක් පැකට්ටුවක් → එක් QR කේතයක් → එක් පරිත්‍යාගයක් → Apeksha රෝහල.

සෑම විකුණුමක්ම රෝහලට ගිය මුදලක් — සහ පාරිභෝගිකයාට පෙනෙන වාර්තාවක්.